

Shahriyar Shawon

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EDUCATION

Drexel University

Bachelor of Science in Computer Science

Philadelphia, PA

September 2020 – June 2025

Concentrations: Operating Systems and Computational Imaging

EXPERIENCE

Susquehanna International Group

Software Engineer Co-op

Bala Cynwyd, PA

September 2021 – March 2024

SFG Team

September 2023 – March 2024

- Enhanced data cohesion and accuracy of stock data by developing an API using C# and .NET to aggregate stock data from 15 MySQL and OracleDB sources
- Decreased redundant data fetching by 93% by redesigning a web app to use a central API
- Reduced stock information lookup time by 95% by introducing a caching mechanism
- Safeguarded privileged data by implementing a role-based access control system

MDF Metadata Team

September 2022 – March 2023

- Improved data accuracy and operational reliability by building an alerting system using Python and PostgreSQL to monitor file delivery integrity and timeliness
- Reduced alert configuration time for developers by 50% by creating a CLI tool that streamlined setup
- Improved a file-user identification tool by implementing multiple detection methods, resulting in more reliable attribution of file usage

CorpSys Team

September 2021 – August 2022

- Enriched an internal project management tool by aggregating developer, stakeholder, pipeline, and environment data to improve team collaboration and project tracking
- Developed the front-end interface for an API to allow general-purpose alerts to be created
- Reduced the use of bridge tables in MySQL by implementing Entity Framework queries, improving query performance by 20%

PROJECTS

Dynamic Image Compression Engine

- Built firmware for two RP2040 chips to interface with FPGA, IO Modules, and Power Monitor Chips
- Implemented custom image format to store lossy and lossless blocks of data using lossy and lossless codecs
- Created a desktop app to control camera settings on the fly and log power/data metrics from hardware
- Engineered a pipeline to compare software-based encoding performance to hardware based encoding

DragonFly3 Auto Connect Script

- Significantly reduced Linux WiFi setup time by developing a one-line Bash script now used by Drexel's College of Computing and Informatics and other Drexel students
- Reduced time to connect to Drexel WiFi by 90% by automating creation of an Enterprise WiFi profile

Drexel Course Information API

- Built a web scraper to parse information from 8,200+ Drexel courses and store them in a SQLite Database
- Created an API to query for classes based on school, subject, credits, and name
- Enabled users to see all prerequisites for classes

SKILLS

Languages: Python, C#, C, C++, Bash/Shell Scripting, HTML/CSS, JavaScript

Databases: MySQL, PostgreSQL, OracleDB, MicrosoftSQL

Frameworks: FastAPI, REST API, Django, Flask, BeautifulSoup, .NET Core, .NET Framework, Entity Framework, jQuery

Tools: Docker, Git, Gitlab CI/CD, Jira, Confluence, CMake, YAML